



SERANGOON JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATIONS 2009

MATHEMATICS
Higher 1

8863

Wednesday
19 Aug 2009

3 hours

Additional materials: Answer paper
Graph Paper
List MF15

INSTRUCTIONS TO CANDIDATES

Write your name, civics group on every answer sheet.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams or graphs.
Do not use staples, paper clips, highlighters, glue or correction fluid.
Write your answers on the separate answer paper provided.

Answer **all** the questions.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

You are expected to use a graphic calculator.

Unsupported answers from a graphic calculator are allowed unless a question specifically states otherwise.

Where unsupported answers from a graphic calculator are not allowed in a question, you are required to present the mathematical steps using mathematical notations and not calculator commands.

You are reminded of the need for clear presentation in your answers.

The number of marks is given in brackets [] at the end of each question or part question.

At the end of the examination, arrange your answers in numerical order.

Attach the cover page on top of your answer script and fasten all your work securely together.

The total marks for this paper is 100.

Section A (Pure Mathematics) 40 Marks

- 1 Solve the simultaneous equations, leaving your answers in exact form.

$$\begin{aligned}x^2 + 3y^2 &= 110 \\x + y &= 12\end{aligned}$$

[4]

- 2 A curve has the equation $y = \ln[x(\sqrt{x-1})]$. If the normal at the point $P(2, \ln 2)$ meets the y -axis at Q , find the exact coordinates of the point Q .

[5]

- 3 Sketch, on a single diagram, for $0^\circ \leq x \leq 360^\circ$, the graphs of $y = \sin x + 1$ and $y = 2 \cos x$.

[2]

Hence, solve the inequalities,

$$\sin x \leq 2 \cos x - 1 \text{ for } 0^\circ \leq x \leq 360^\circ \text{ and}$$

[2]

$$\sin(\theta + 90^\circ) \leq 2 \cos(\theta + 90^\circ) - 1 \text{ for } -90^\circ \leq \theta \leq 270^\circ.$$

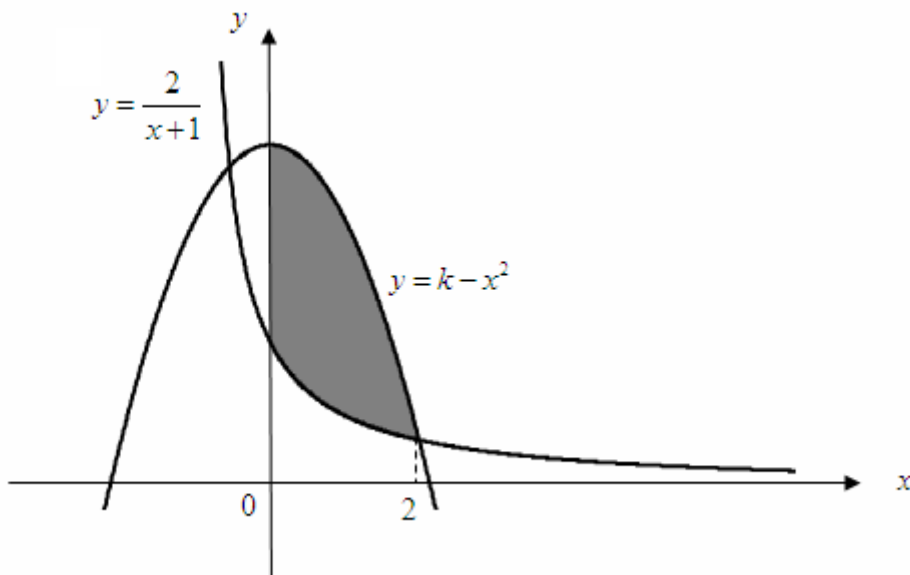
[2]

- 4 (a) The gradient at any point on a particular curve is given by $\frac{dy}{dx} = (5 - 2x)^2$. Given that the curve passes through the point $(3, -1)$, find the equation of the curve.

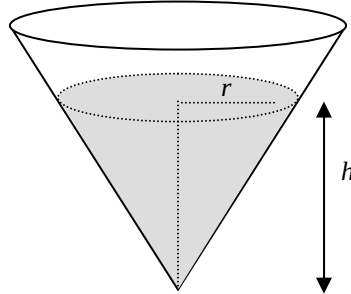
[3]

- (b) The diagram below shows part of the curves $y = \frac{2}{x+1}$ and $y = k - x^2$. Given that the shaded area is $(4 - 2 \ln 3)$ units², find the value of k .

[5]



- 5 (a)** Water is poured into a small cone at a rate of $5\text{cm}^3/\text{s}$. Given that the height of the water is twice as much as the radius of the surface area of the water at any instant, find the exact rate of increase in radius of the surface of the water when the height is 3 cm. [3]



- (b)** The equation of a curve is given as $y = kx^3 - ax + b$ where a, b and k are non zero real numbers and $k > 0$. It is given that the curve has a turning point at $(2,3)$,

- (i)** [2]

how that $a = 12k$ and $b = 3 + 16k$.

- (ii)** Find the x -coordinate of the other turning point and state clearly with reasons, the nature of both the turning points. [3]

- 6** The functions f and g are defined as follows:

$$f : x \mapsto x^2 - 3, \quad x \in \mathbb{R}, \quad x \geq 0$$

$$g : x \mapsto e^x, \quad x \in \mathbb{R}.$$

- (i)** Explain why the inverse function f^{-1} exists. Hence, define f^{-1} in similar form. [4]
- (ii)** Explain why fg exists and sketch the graph of $y = fg(x)$, showing clearly any asymptotes and the axial intercepts. State the domain of fg . [4]
- (iii)** Find $g^2(0)$. [1]

[Turn over

Section B (Statistics) 60 Marks

- 7 (a) (i) Explain briefly what is meant by the term “ Systematic Sampling”. [1]
- (ii) Explain how a sample of 30 students can be selected from a student population consisting of 1500 students by a systematic sampling. [2]
- (b) A sample of 200 observations of a random variable X is taken from a large population as shown in the table below:

x	3	4	5	6	7	8	9	10	11	12
Frequency	12	23	30	25	33	25	20	19	8	5

Find the unbiased estimates of the population mean and the population variance correct to two decimal places.

Hence find an approximate value of the probability that the sample mean lies between 5 and 7. [4]

- 8 A company manufactures computer chips, some of which are defective. It is known that the proportion of defective chips is 15%.

- (i) A random sample of 20 chips is examined. Find the probability that there are at least 4 defective chips. [2]
- (ii) A random sample of n chips is examined. Find the smallest possible value of n such that the probability of this sample having no defective chips is less than 3%. [2]
- (iii) Chips are packed in boxes where each box contains 20 chips. 100 such boxes are examined. Using a suitable approximation, find the probability that at most 60 boxes contain less than 4 defective chips [4]

- 9 A fruit stall sells 2 types of melons: A and B. The masses of Type A melons and

Type *B* melons are assumed to have independent normal distributions with means and variances as shown in the table.

Melons	Mean (kg)	Variance (kg ²)
Type A	2.5	0.7
Type B	1.3	0.3

Type A melons are sold at \$0.50 per kg and Type B melons at \$1.20 per kg.

- (i) ind the probability that a randomly chosen Type A melon has selling price exceeding \$1.50 and a randomly chosen Type B melon has a selling price exceeding \$2. [4]
- (ii) ind the probability that the total selling price of a randomly chosen Type A melon and a randomly chosen Type B melon is more than \$3. [3]
- (iii) ind the probability that the difference in the selling price of a randomly chosen Type A melon and a randomly chosen Type B melon is within $\pm \$0.50$. [3]

- 10 A random sample of 150 batteries is tested and the lifetime, x hours, of each battery is measured. The results are summarized by

$$\sum x = 24\,236, \sum x^2 = 4\,212\,818$$

Of the 150 batteries, 60 batteries of a particular brand are removed from the sample. These 60 batteries are then tested and the lifetime, y hours, of each battery is measured, with results summarized by

$$\sum y = 6\,746, \sum y^2 = 778\,765$$

The population mean lifetime is denoted by μ hours. The null hypothesis $\mu = 200$ is to be tested against the alternative hypothesis $\mu < 200$.

- (i) Using the **remaining batteries** as the sample, carry out a test, at 1% significance level, of the same null and alternative hypotheses. [8]
- (ii) Explain **p – value** and “**1% level of significance**” in the context of question. [2]

- 11 An instrument used to measure the levels of cholesterol levels in a person’s blood is monitored on a random sample of 10 people. The levels are also measured using an **accurate** laboratory procedure. The following table gives the instrument measure, x , and the laboratory procedure measure, y , in suitable units.

x	231	156	190	242	212	180	251	320	239	172
y	229	153	192	244	216	175	254	110	235	175

- (i) Which of the regression lines, y on x or x on y , should be used to express an approximate linear relation between x and y ? Give a reason for your choice. [2]
- (ii) Sketch a scatter diagram for the data. [2]
- (iii) Calculate the product moment correlation coefficient for the data and comment on this value. [2]
- (iv) It was suspected that a data pair was recorded wrongly. With reference to the scatter diagram in (ii), identify the suspect data pair. [1]

[Turn over

- (v) Remove the suspect data pair and recalculate the product moment correlation coefficient and the equation of a suitable least squares regression line chosen in part (i). [2]
- (vi) Using the equation of a suitable least squares regression line, estimate the value of instrument measure when the laboratory procedure measure is 210. Leave your answer to the nearest 1 decimal place. [1]
- (vii) Comment on the reliability of the estimated value of the instrument measure obtained in (vi). [2]

12 (a) Events X and Y are such that $P(X) = \frac{13}{20}$ and $P(Y) = \frac{3}{10}$. Find $P(X \cap Y)$ when X and Y are

- (i) mutually exclusive, [1]
- (ii) independent. [1]

Hence, find $P(Y|X')$ for each of the cases above where X' denotes the complement of the event X . [3]

(b) In a particular junior college, the student population is divided in the ratio 4:5 between the girls and boys in that order. 20% of the girls and 15% of the boys wear glasses.

- (i) Using a tree diagram, find the probability that a student chosen at random wears glasses. [2]

- (ii) If a student chosen at random wears glasses, find the probability that the student is a girl. [3]
- (c) A game of an unbiased die is played between two players A and B as follows:
If a player throws a '1', he throws the die again. Otherwise he passes the die to the other player. Whoever throws a '1' consecutively two times, wins the game and the game ends there.
Player A throws the die first. Find the probability that the game ends after 5 throws. [3]

End of Paper